University ERP System — Project Report

1. Introduction

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The University ERP System is a desktop-based academic management solution designed to streamline operational activities across multiple user roles within a university environment. Developed as a modern and efficient desktop application, the system provides a clean and accessible graphical user interface (GUI) that follows role-based access principles. Each user type—Student, Instructor, and Administrator—interacts only with the features necessary for their academic responsibilities, ensuring clarity, usability, and security.

Students use the platform to navigate their academic journey efficiently. They can browse the complete course catalog, register or drop course sections depending on institutional deadlines, view personalized timetables, and download academic transcripts. Instructors manage their assigned course sections, create dynamic assessment schemas by defining weightage distributions for various components, enter student grades, and monitor live class statistics for continuous evaluation. Administrators perform institutional-level operations including creating and managing user accounts, configuring course offerings, scheduling sections, and managing system-wide controls such as the platform's Maintenance Mode.

Technically, the application is built using Java 21, taking advantage of modern syntax improvements provided through Preview features. The user interface is implemented using Java Swing styled by FlatLaf, offering a contemporary Light/Dark theme experience. The persistence layer relies on JDBC and connects to MySQL via HikariCP, ensuring high-performance database access. Security is enforced through a dedicated "Shadow Password"

subsystem, providing strong credential protection and isolation between authentication and operational data.

2. Database Design

2.1 Separation of Concerns Architecture

To maintain a secure and maintainable system, the ERP adopts a Separation of Concerns architecture at the database level. Two physically separated databases are used—one dedicated to authentication and security, and the other to operational academic data. These databases connect logically through the user_id foreign key. This division ensures that sensitive login credentials remain isolated from other academic records, reducing attack surface and improving modularity during development.

2.2 Database 1: auth_db (Authentication & Security)

Table: users_auth

Purpose:

Stores user credentials and authentication-related metadata.

Columns:

user_id — Primary Key (Auto-Increment)

username — Unique login identifier

role — Enum: STUDENT, INSTRUCTOR, ADMIN

password_hash — BCrypt hashed password

status — ACTIVE / LOCKED

last_login — Timestamp of most recent login

failed_attempts — Number of consecutive failed attempts

lockout_end — Timestamp until which login is disabled

This table forms the core security layer, supporting lockout protection and secure password storage.

2.3 Database 2: erp_db (Operational Data)

Table: students

Columns: user_id (FK), roll_number, program, year

Table: instructors

Columns: user_id (FK), department, title

Table: courses

Purpose: Master catalog of subjects

Columns: course_code (PK), title, credits

Table: sections

Purpose: Specific course offerings for a semester

Columns: section_id (PK), course_code, instructor_id, day_time, room, capacity, semester, year

Table: enrollments

Purpose: Maps Students to Sections

Columns: enrollment_id (PK), student_id, section_id, status (REGISTERED/DROPPED), final_grade

Table: assessment_schema

Purpose: Stores weightage rules for a section

Columns: schema_id, section_id, component_name, max_marks, weightage

Table: grades

Purpose: Stores individual assessment scores

Columns: grade_id, enrollment_id, component_name, score

Table: settings

Purpose: Global key–value configuration store

Columns: setting_key, setting_value

3. Implementation Details

3.1 Authentication & Security (Shadow Password Model)

The application enforces strong security through a "Shadow" password architecture. User passwords are never stored in plain text; instead, the system uses BCrypt with a Work Factor of 12, leveraging the jBCrypt library. This ensures that each password hash is computationally expensive to crack and includes a unique salt.

To prevent brute-force login attacks, the AuthService maintains a counter for failed attempts. After five consecutive failures, the account is temporarily locked. The system computes a lockout expiration time using:

lockout_end = current_time + 2 minutes

During this lockout period, the user cannot authenticate, and the LOCKED status is enforced until the timestamp expires.

3.2 Maintenance Mode Logic

A system-wide "Maintenance Mode" serves as an administrative control to temporarily freeze critical write operations. This mode is stored as a boolean flag in the settings table.

The AccessService functions as a centralized interception layer. Before any write operation (e.g., registration, dropping a course, entering grades), the service checks this flag. If Maintenance Mode is active:

Write operations are blocked

A warning exception is raised

The GUI displays a high-visibility Yellow Maintenance Banner via the MaintenanceBanner component

All input forms are disabled for non-admin users

3.3 Grading Algorithm

Instructors define section-specific assessment schemas, allowing dynamic weight distribution across assessment components (e.g., Quizzes, Midterm, Final Exam).

For each enrolled student, the InstructorService computes the final percentage using:

Weighted Score = (Score Obtained / Max Marks) × Weightage%

Final Percentage = Sum of all Weighted Scores

After computing the percentage, the system generates a letter grade following rules such as:

95% and above → A+

90–94% → A

…

Below 30% → F

The resulting letter grade is stored in the enrollments.final_grade column.

4. Code Structure

The project follows a well-defined Layered Architecture, ensuring modularity, maintainability, and testability.

4.1 edu.univ.erp.ui — Presentation Layer

Built with Java Swing components (JPanels, JTables, JDialogs)

Utilizes MigLayout for responsive forms

Styled with FlatLaf for Light/Dark theme support

No direct database access occurs in this layer

4.2 edu.univ.erp.service — Business Logic Layer

Contains core application logic

Services such as StudentService, InstructorService, AdminService

Enforces constraints: capacity checks, registration deadlines, grading weight validation

4.3 edu.univ.erp.data — Data Access Layer (DAO)

Manages all JDBC operations

Uses HikariCP connection pooling

Executes raw SQL queries

Maps ResultSet rows into domain objects

4.4 edu.univ.erp.auth — Authentication Module

Manages login, logout, UserSession tracking

Enforces lockout logic

Bridges the auth database with the UI layer

4.5 edu.univ.erp.domain — Domain / DTO Layer

Contains Plain Old Java Objects (POJOs)

Used for transferring data between the DAO and Service layers

Represents entities such as Student, Course, Section, Grade, etc.

5. Extras & Bonus Features

The system implements several enhancements that improve usability, performance, and output quality:

5.1 CSV Export (OpenCSV)

Students can export academic transcripts, and instructors can download gradebooks for analysis. Uses the OpenCSV library for correctly formatted and encoded CSV files.

5.2 Background Processing (SwingWorker)

Long-running tasks—such as loading large catalogs or computing class statistics—run in background threads, ensuring the UI does not freeze.

5.3 Robust Validation Framework

Ensures assessment schema weights total exactly 100%

Prevents duplicate enrollment attempts

Validates course capacity constraints

Guards against invalid data entry at the UI and service layers

6. Screenshot Placeholders

Figure 1: Login Screen

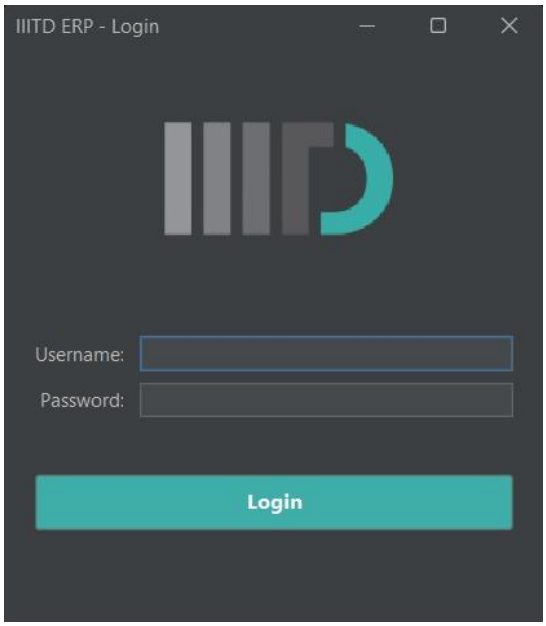


Figure 2: Admin Dashboard

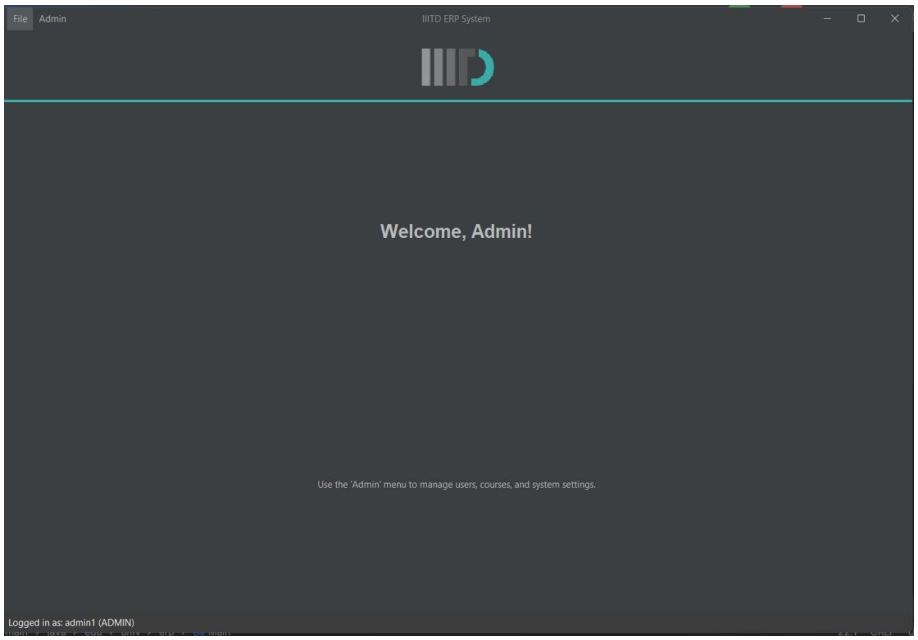


Figure 3: Create users

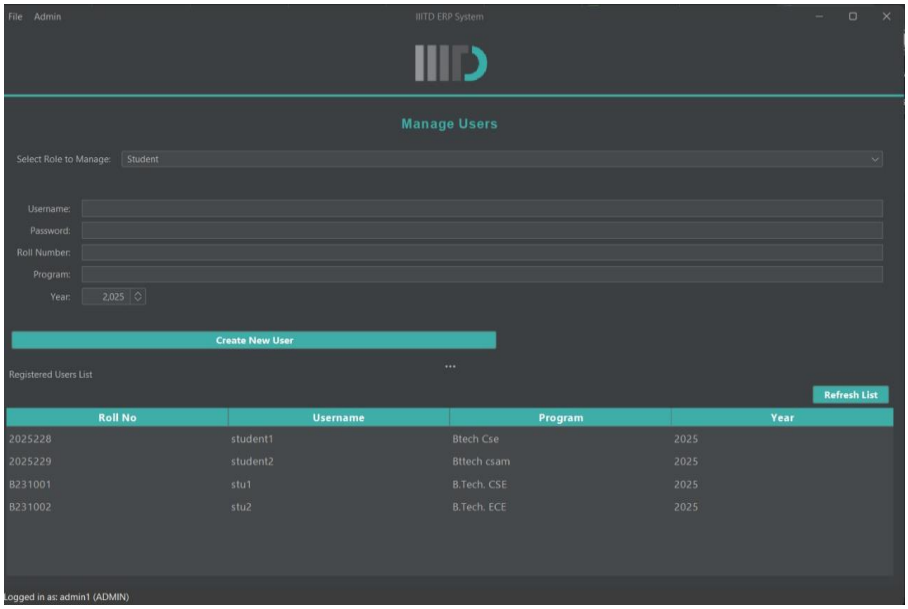
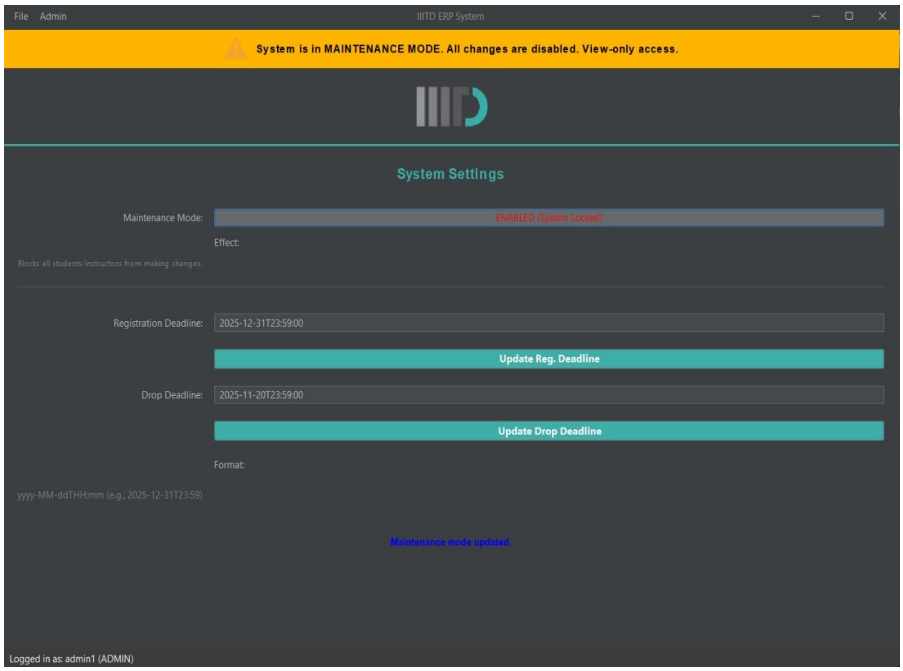


Figure 4: Instructor Gradebook



Figure 5: Maintenance Mode View


File Admin
IIITD ERP System
System is in MAINTENANCE MODE. All changes are disabled. View-only access.
System Settings
Maintenance Mode:
ENABLED (System Locked)
Effect:
Blocks all students/instructors from making changes.
Registration Deadline:
2025-12-31T23:59:00
Update Reg. Deadline
Drop Deadline:
2025-11-20T23:59:00
Update Drop Deadline
Format:
yyyy-MM-ddTHH:mm (e.g., 2025-12-31T23:59)
Maintenance mode updated.
Logged in as: admin1 (ADMIN)